Caymann Velingkar

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Profile Summary

Technology enthusiast with a background in software engineering, AI, machine learning and data analytics. Experienced in developing applications, building intelligent solutions, automating workflows and solving technical problems. Thrives in collaborative environments with a solution-oriented mindset, systems thinking and a commitment to continuous learning.

Experiences

OneShield Softwares | Quality Assurance Intern | *Certificate* *Jul 2023 – Sep 2023*

- Developed structured test workflows using the Behavior-Driven Development (BDD) framework and conducted detailed validation of 50+ insurance process fields, ensuring accuracy and minimising errors across diverse data types.
- Automated 25+ test scenarios with Selenium, improving accuracy and consistency in software functionality checks.
- Refactored test processes and collaborated with a team of 15 members to enhance efficiency while maintaining high-quality standards across diverse data forms.

Cursor Council | Magazine Secretary | *Certificate* *Sep 2022 – Jun 2023*

- Tailored an interactive college department magazine with a companion app that scanned pages for clues, engaging readers with treasure hunts and puzzles thus boosting user engagement and publication visibility.
- Designed and managed online forms for talent management and content coordination, streamlining data collection and allocation of more than 250 participants based on interests.
- Spearheaded poster and creative campaigns, boosting engagement and initiative visibility.
- Coordinated 10+ council activities through structured workflows and meeting minutes, liaising with internal and external bodies to ensure timely execution.

Education

MSc Advanced Computing | King's College London *Sep 2025 – Sep 2026*

BEng in Computer Engineering | Goa Engineering College, India | CGPA: 8.92/10 *Nov 2020 – Aug 2024*

Technical Skills

- **Computer Fundamentals:** Data Structures & Algorithms, Computer Networks, Software Architecture
- **Programming & Frameworks:** Python, C++, JavaScript, Flutter, PHP, Object Oriented Programming
- **Data Analytics:** Excel, Feature Engineering, MySQL, Exploratory Data Analysis, Statistical Analysis, Power BI
- **Data Science:** Machine Learning, Image Processing, Deep Learning, NLP, Transformers, Generative AI
- **Project Management:** Agile, Scrum, Waterfall, Project Planning, Documentation
- **Design & Tools :** Canva, Figma, LaTeX, Jira, Linux, GitHub, Windows, MS Office

Projects

AI-Powered Semantic Search Photo Gallery | *Link*

- Developed a multimodal (text, image, audio) semantic search engine using CLIP embeddings and FAISS search.
- Achieved top-30 retrieval across 10,000+ images with an average query latency of ~ 2.17 milliseconds, significantly reducing user effort and time spent searching for specific photos.
- Implemented automatic preprocessing, incremental embedding updates, and metadata extraction (EXIF & geolocation) stored in SQLite, coupled with a real-time Watchdog pipeline for continuous ingestion.
- Created hierarchical (Agglomerative) album clustering and an interactive Flask-based dashboard, providing image organization, filtering, visualization, and CRUD operations.

Immersive Spatial Audio Powered by Computer Vision | *Link*

- Architected a Flutter mobile app with a distributed backend using FastAPI and Redis Streams for asynchronous communication between audio and video processing units, ensuring real-time performance and system resilience.
- Integrated a multi-camera stereo vision system (two cameras) with Mediapipe & OpenCV, performing real-time head-pose estimation via 3D triangulation.
- Crafted an adaptive binaural audio pipeline using individualised HRIRs from the LISTEN dataset, convolved via FFT for efficient real-time spatialisation. Simulated realistic acoustic environments using Room Impulse Response (RIR) and RT60 modeling with Pyroomacoustics, enhancing immersive audio perception.

Achievements

- **NHS Clinical AI & Quantum Hackathon, London:** Grand Final: *1st Place* | Qualifier: *1st Place*
- **Music Technology Hackathon:** Queen Mary University of London: *3rd Place*
- **Hackathon:** Indian Institute of Technology Goa - Cepheus: *1st Place*
- **Computing in Python:** NPTEL Swayam: *Top 1% nationally with 93% score*